Early Prosodic Experience Has a Lasting Effect on Adult Prosody Processing

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Introduction. Prosody is among the earliest learned components of language. Rhythmic and intonational cues reach the fetus before birth[1], neonates prefer prosodically familiar speech[2], and early prosodic structure supports foundational processes including segmentation[3], lexical acquisition[4], and syntactic bootstrapping.[5] Because prosody is acquired early, suprasegmental properties of a parent's native language could shape prosodic processing of children who ultimately grow up monolingual in the societal language. This study investigates whether the rhythmic and tonal characteristics of parents' native languages affect prosody comprehension into adulthood, even for individuals who report English as the only language they learned in childhood.

Method. We tested the prosody comprehension of 362 college students who self-identified as monolingual English speakers. Of these, 37% had two native English-speaking parents and 47% had no native English-speaking parents. Participants' parents' languages were classified by (i) rhythmic type (English stress-timed; non-English stress-timed; syllable/mora-timed) and (ii) tonality (English; non-tonal non-English; tonal). Participants were tested on their ability to discriminate prosodic form and to comprehend affective prosody, declarative/question prosody (turn-end), lexical stress, phrase stress, phrase boundary prosody, and contrastive stress.[6] Bayesian ANOVAs were used to assess how the rhythmic and tonal type of parents' languages affected participants' accuracy and RTs.

Results. Parental *rhythmic type* robustly affected participants' RTs but not accuracy. Participants whose mothers spoke syllable-timed languages were decisively slower overall than those with English-speaking mothers (BF = 2.1×10^{3}; Fig. 1A), with contrastive stress (post hoc BF = 4.8; Fig. 1C) being most affected. Paternal rhythmic type also influenced RTs: syllable-timed paternal input was associated with longer RTs compared to English (BF = 2.2×10^{3}) or non-English stress-timed paternal input (BF = 8.6×10^{10}; Fig. 1B). Parental *tonal type* did not affect accuracy, with one exception: participants whose mothers spoke tonal languages performed worse on contrastive stress than those with English-speaking mothers (BF = 150.4) or mothers who spoke a non-tonal language other than English (BF = 23.8). Parental language tonality had a greater impact on RTs: participants whose mothers spoke tonal languages were decisively slower overall than those with English-speaking mothers (BF = 409.3; Fig. 1E). This was driven primarily by lexical stress (post hoc BF = 318.8; Fig. 1G). Paternal tonality had a greater impact than maternal tonality. Overall, participants with fathers who spoke tonal languages were decisively slower than those whose fathers spoke English (BF = 4.0×10^{23}) or another non-tonal language (BF = 3.6×10^{21}; Fig. 1F). Post hocs revealed this was true for declarative/question prosody, lexical stress, phrase stress, phrase boundary prosody, and contrastive stress (Fig. 1H).

Discussion. Across rhythmic and tonal dimensions, these analyses suggest that prosodic comprehension in adulthood is shaped by early exposure to the prosodic structure of parents' native languages. Critically, the effects were greater for RT than accuracy, indicating that these listeners almost always understood the prosodic distinctions but differed in the processing effort required to compute them. From a theoretical perspective, these findings support accounts in which prosody is a modular but experience-sensitive system, with processing patterns that are tuned early in development and remain stable over time. Early exposure to non-English rhythmic or tonal prosody may calibrate the predictive expectations listeners bring to suprasegmental parsing, such that English cues mismatch early-established priors and lead to slower integration. Taken together, these findings suggest that the adult prosodic system retains a durable imprint of early perceptual experience, even after decades of English-dominant exposure. More broadly, the results challenge the assumption that adults who speak only English form a prosodically homogeneous group. They also reinforce the view that early prosodic input is a lasting, structured source of individual variation that sentence processing models should explicitly incorporate.

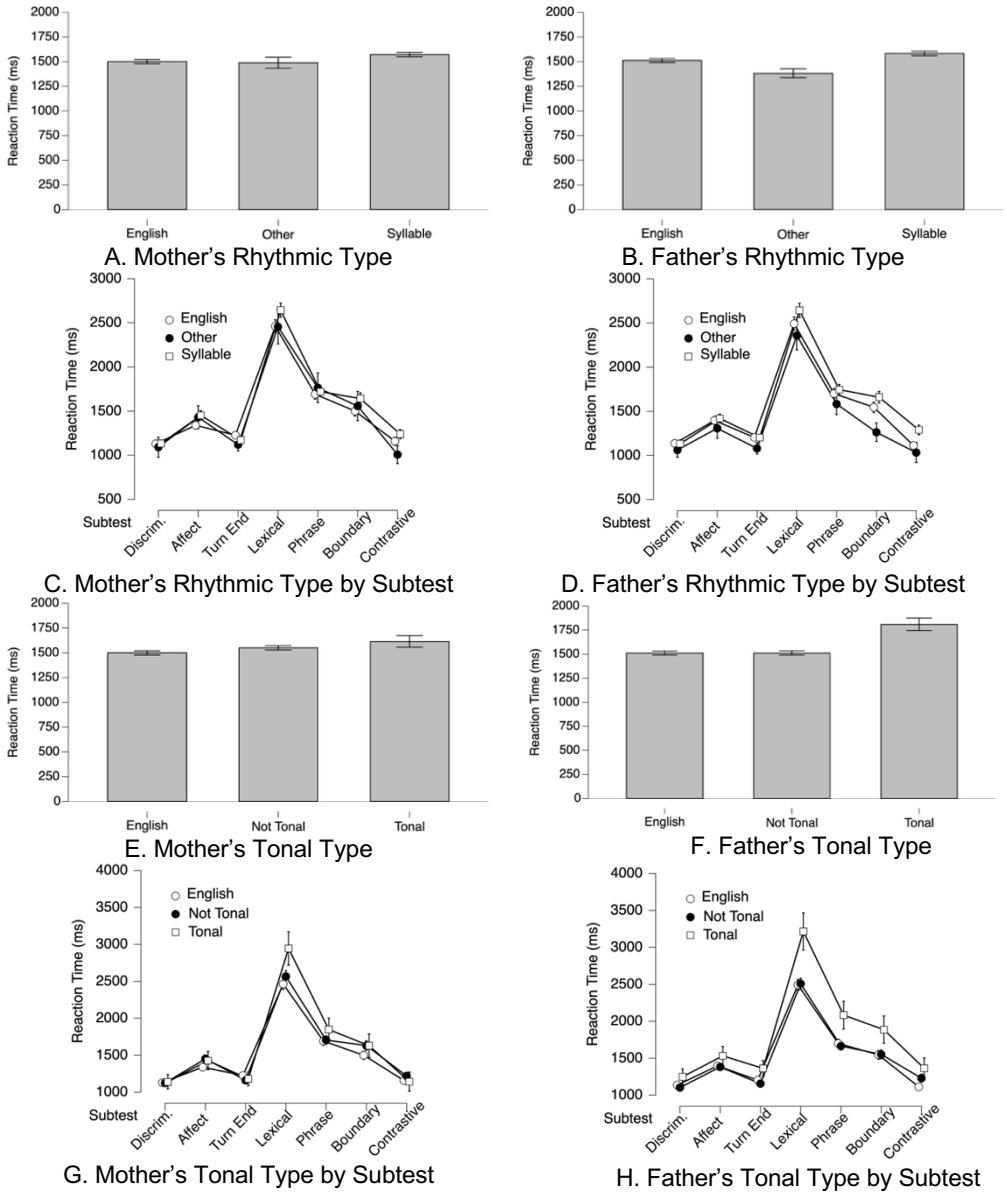

Figure 1. Error bars are 95% credible intervals

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